CHOLESTEROL SYNTHESIS AND DEGRADATION

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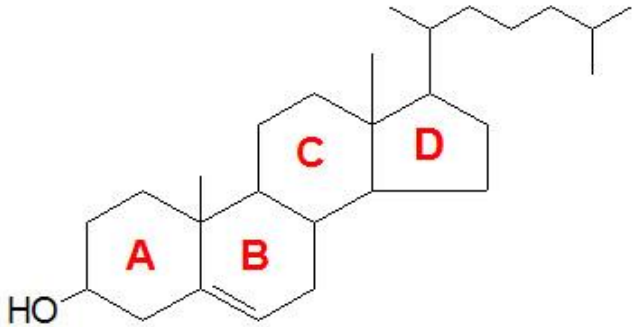
INTRODUCTION

- Cholesterol roles
  - membrane structure
  - precursor for the synthesis of the steroid hormones, bile acids and and vitamin D.
- Both dietary cholesterol and that synthesized *de novo* are transported through the circulation in lipoprotein particles. The same is true of cholesteryl esters, the form in which cholesterol is stored in cells.
- The synthesis and utilization of cholesterol is tightly regulated in order to prevent over-accumulation and abnormal deposition within the body



Cholesterol molecule

- The typical daily Western diet contains approximately 500 mg (1.2 mmol) of cholesterol daily, mainly in meat, eggs, and dairy products
- Total plasma cholesterol is about 5.2 mmol/L
- Under normal circumstances, 30-60% of this is absorbed from the gut (About half the cholesterol arises by biosynthesis and the remainder from diet)
- Human beings synthesize 1g cholesterol each day, mainly in the liver. The rate of its endogenous synthesis is determined by dietary intake. For this reason both dietary intake and biosynthesis are important in determining its plasma concentration.

- Cholesterol has a low solubility in water.
- Only about 30% of circulating cholesterol occurs in the free form, the majority is esterified through the hydroxyl group to a wide range of long-chain fatty acids including oleic and linoleic acids.
- Cholesterol esters are even less soluble in water than free cholesterol.
- Present in plasma in relatively high amounts for the poor solubility because of the presence of a range of lipoproteins which incorporate and thereby solubilize the cholesterol molecule
- Within these lipoproteins, the hydrophobic cholesterol esters are located in the core of the molecule, with free cholesterol in the outside layer.
- The dietary cholesterol brought to the liver is mostly in the form of free cholesterol.
- The cholesterol present in VLDL and LDL is mostly in the form of cholesteryl esters. Esters are also the tissue storage form of cholesterol (they are stored in lipid droplets).
  - In the plasma, cholesterol is esterified by the enzyme lecithin cholesterol acyltransferase
  - and in the cells cholesterol is esterified by the acyl-CoA: cholesterol acyltransferase.

Chylomicron metabolism

- After absorption, triglycerides and cholesterol are transported to the peripheral tissues in the form of chylomicrons (have apoproteins B 48, E and apoC-II).
  - In the peripheral tissues chylomicrons are hydrolyzed by lipoprotein lipases (activated by apoC-II) on capillary surfaces
    - Fatty acids are used as fuel or precursors of endogenously synthesized fats
    - apoC-II returned to HDL
  - Cholesterol rich residues, the chylomicrom remnants are taken up by the liver after binding to specific receptors via apoE component

VLDL, IDL and LDL metabolism

- Triglycerides and cholesterol in excess of the livers needs are exported into blood as VLDL which have apoproteins B 100, apoE and apoCs

  (apoE and apoCs are picked from HDL)
  - Triglycerides in VLDL are hydrolyzed by lipoprotein lipases (activated by ApoC-II on capillary surfaces to form IDL which are rich in cholesterol
    - Half of IDL taken up by the liver
    - Half of IDL converted to LDL (the major carrier of cholesterol in blood)
      - LDL carry cholesterol to peripheral tissues and regulate denovo cholesterol synthesis there
- VLDL remnants and LDL deliver cholesterol to tissues by binding to the apoB/E receptor (LDL receptor).

- High levels of **LDL cholesterol** are directly correlated with increased risk of cardiovascular disease. Thus, LDL cholesterol is known as “bad cholesterol”.

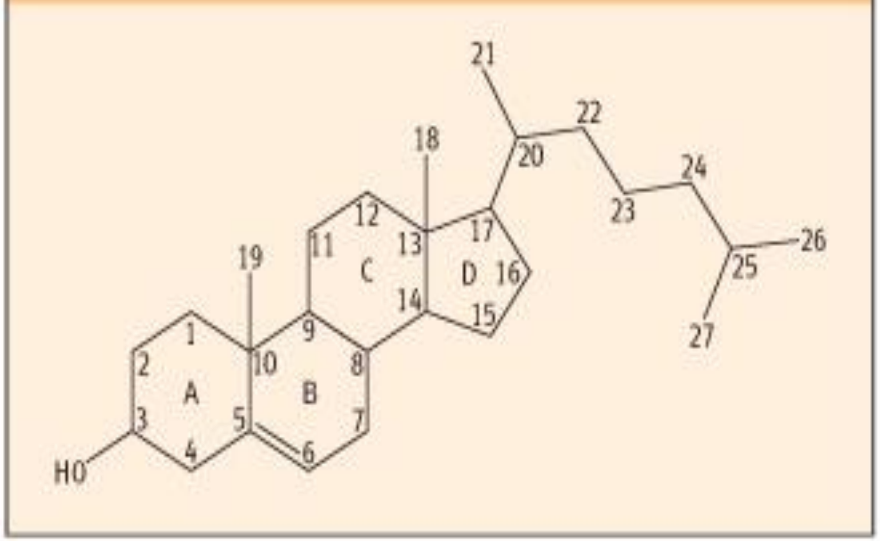
METABOLISM OF HDL

- HDL protein components are synthesized in the liver and intestine
- HDL particles are a reservoir for apolipoproteins (especially **apoCII** and **apoE**).
- Small nascent HDL can diffuse close to peripheral cell membranes where HDL picks up cholesterol.
  - Cholesterol efflux regulatory protein (CERP) plays a major role in cholesterol efflux from cells
- Back in circulation **lecithin cholesterol acyltransferase (LCAT)** converts cholesterol to CE on HDL.
- This reaction is activated by apoA-I
- HDL bind to **Scavenger receptor class B member 1** (SRB1) receptors on adrenal cells, gonads, and the liver, and delivers cholesterol esters for **synthesis** of **steroid** hormones and bile salts for excretion
- The reverse cholesterol pathway is the transport of cholesterol from peripheral cells to the sites of **synthesis** or **excretion.**
- **HDL cholesterol** levels are inversely related to risk of cardiovascular disease. Therefore, HDL has been called “good cholesterol”.

Cholesterol biosynthesis

- Cholesterol synthesis occurs in the cytoplasm and microsomes from the two-carbon acetate group of acetyl-CoA.
- The process has four major steps:
  1. Acetyl-CoAs are converted to mevalonate
  2. Mevalonate is converted into activated isoprene units
  3. Polymerization of six 5-carbon isoprene units to form a 30 carbon linear structure of squalene
  4. Stage 4, the cyclization of squalene forms the four rings of the steroid nucleus, and a further series of changes (oxidations, removal or migration of methyl groups) leads to the final product cholesterol
- Biosynthesis of cholesterol (many C-C and C-H bonds) requires:
  - Source of carbon atoms. Acetyl CoA provides a high energy starting point
  - Reducing power provided by NADPH from the pentose phosphate pathway
  - Additional energy provided by the breakdown of adenosine triphosphate (ATP)
- Cholesterol is not required in mammalian diet because all cells can synthesize it from simple precursors
- All tissues with nucleated cells are capable of cholesterol synthesis, which occurs in the endoplasmic reticulum and cytosol.
- All the biosynthetic reactions occur within the cytoplasm
- Overall the production of 1 mole of cholesterol requires 18 moles of acetyl-CoA, 36 moles of ATP and 16 moles of NADPH.

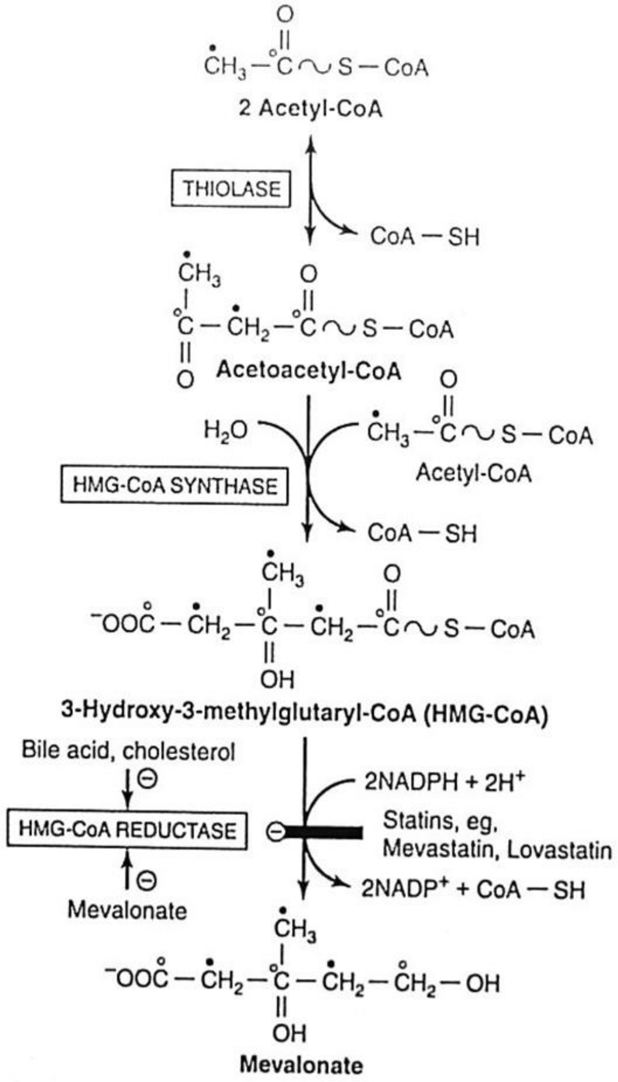
Structure of cholesterol



- 27 carbon
- 2 angular methyl groups
- Solitary hydroxyl group attached to carbon atom 3
- One double bond between carbon atoms 5 and 6
- Three-dimensional terms the ring structure of cholesterol is approximately planar

Cholesterol is made from acetyl-CoA in four stages

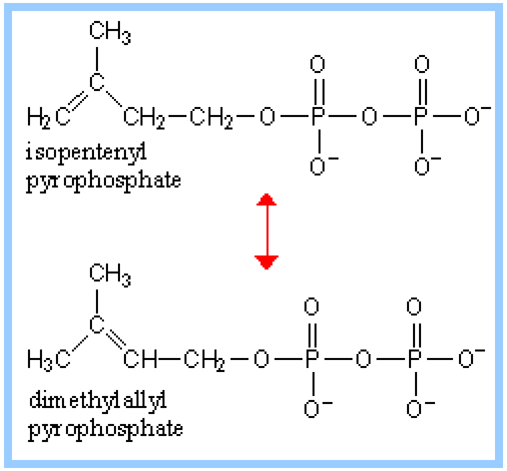
- In stage 1, three acetate units condense to form a six carbon intermediate, mevalonate

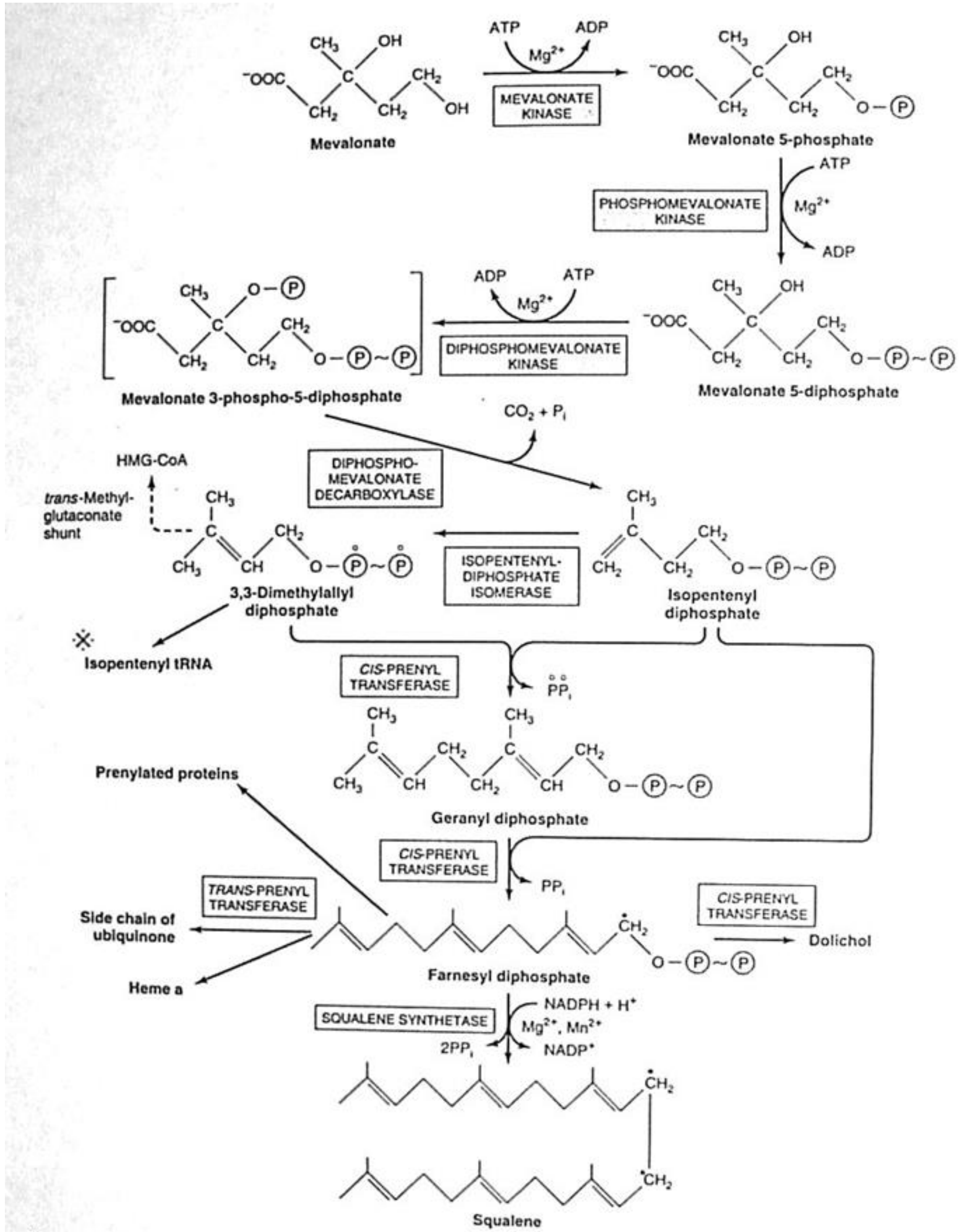


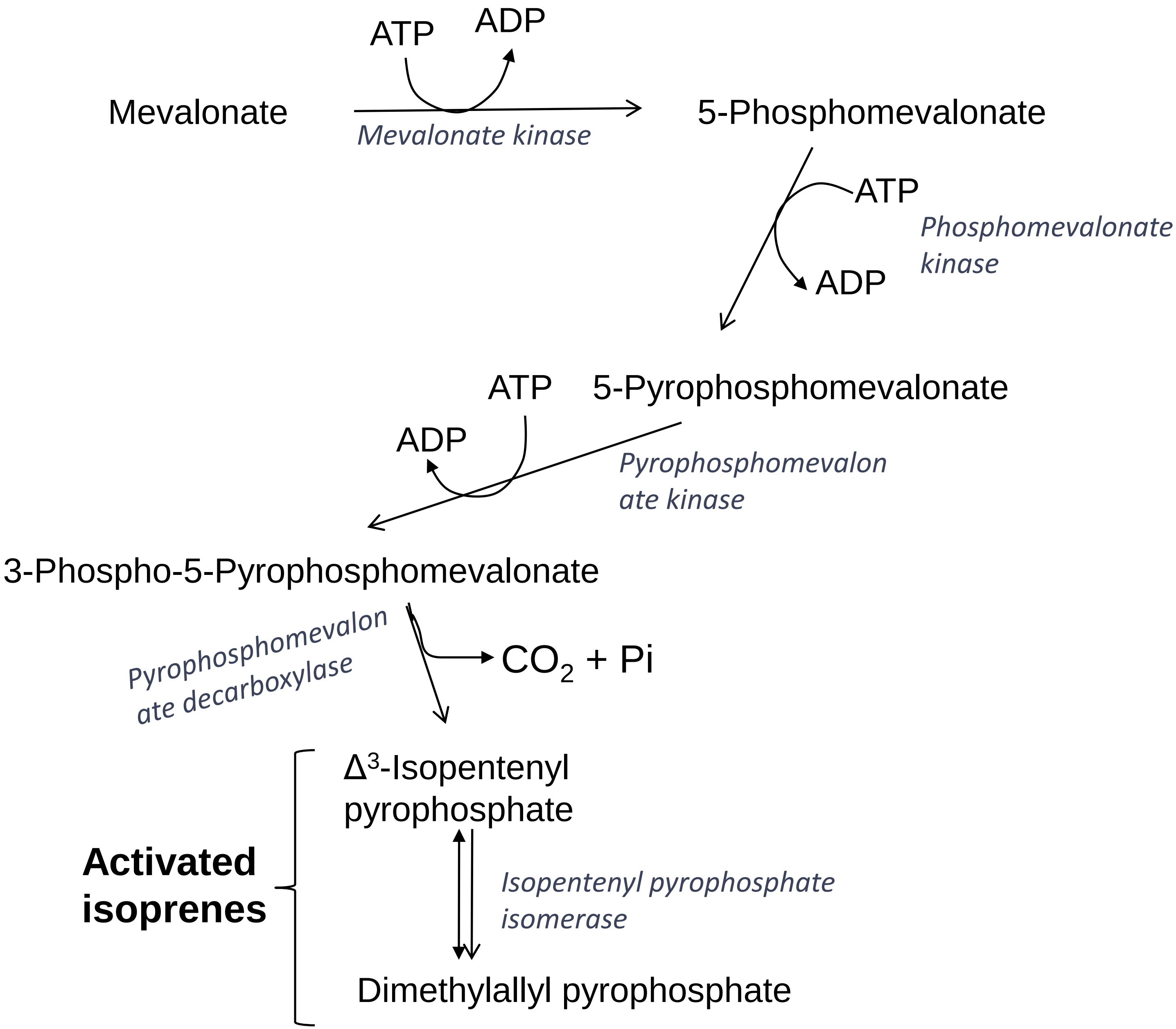
- The first two reactions catalyzed by acetoacetyl-CoA thiolase and HMG-CoA synthase respectively are reversible and do not commit the cell to the synthesis of cholesterol
- The cytosolic HMG-CoA synthase is dictinct from the mitochondrial isoenzyme that catalyzes HMG-CoA synthesis in ketone body formation
- The third reaction is the committed step: the reduction of HMG-CoA to mevalonate catalyzed by HMG-CoA reductase
  - Major point of regulation of cholesterol synthesis and is is the site of action of most effective class of cholesterol –lowering drugs the *HMG-CoA reductase* inhibitors (statins)
    - The inhibition of HMG-CoA reductase activity leads to the lowering of intracellular cholesterol concentration, to a consequent upregulation of the apo B/E receptor and, through that route, to the lowering of the plasma cholesterol concentration.
  - The enzyme is active in a non phosphorylated state and is inactive when phosphorylated
  - Two molecules of NADPH each donate two electrons

Stage 2 involves conversion of mevalonate into activated isoprene units

- In the next stage of cholesterol synthesis, three phosphate groups are transferred from three ATP molecules to mevalonate- activates mevalonate and increases its solubility
- Mevalonate is phosphorylated sequentially by three kinases and after decarboxylation isopentenylpyrophosphate is formed
- Isomerization of isopentenylpyrophosphate yields the second activated isoprene unit, 3, 3 dimethylally pyrophosphate

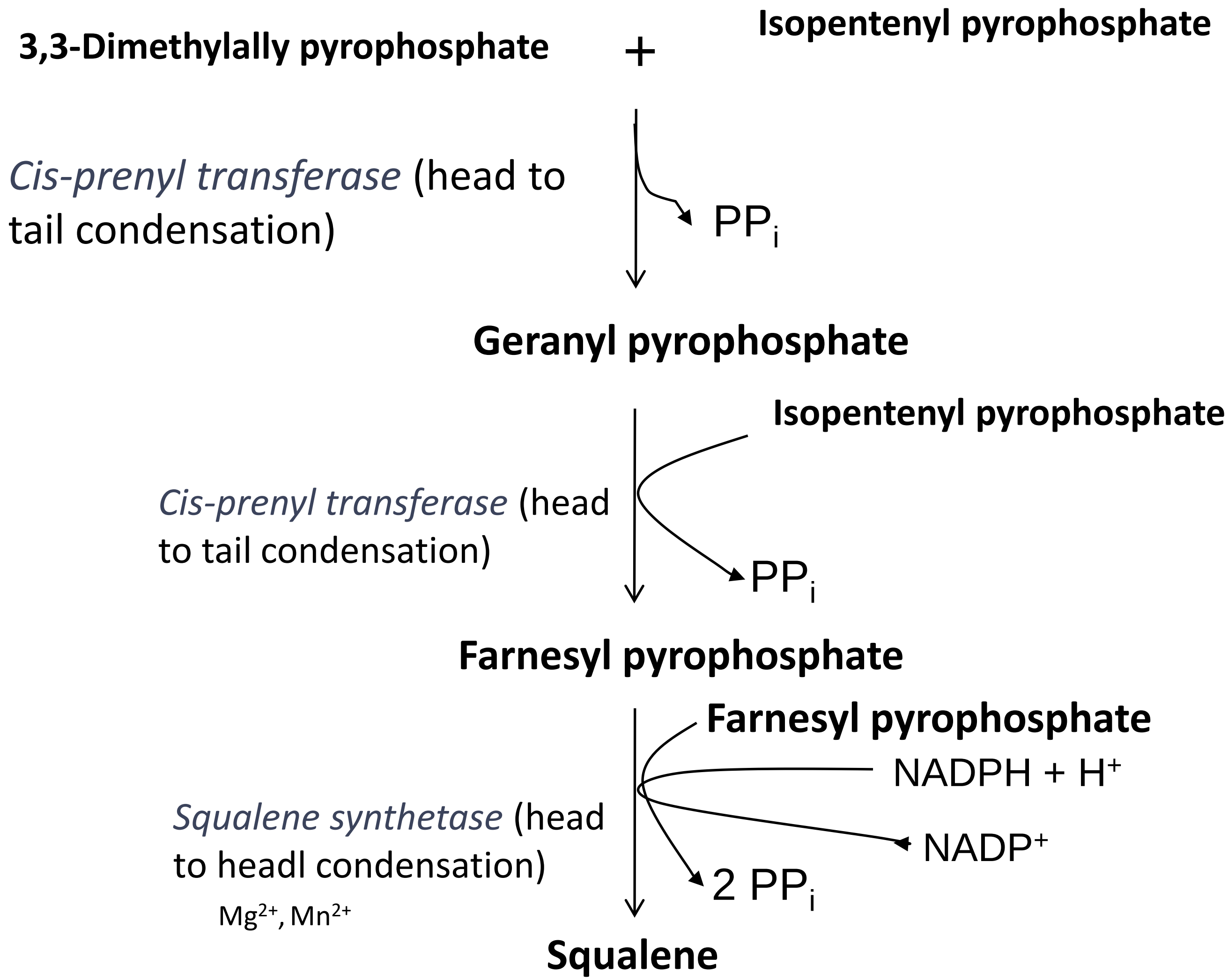



Mevalonate
ATP
ADP
Mg^{2+}
MEVALONATE KINASE
Mevalonate 5-phosphate
PHOSPHOMEVALONATE KINASE
DIPHOSPHOMEVALONATE KINASE
Mevalonate 5-diphosphate
Mevalonate 3-phospho-5-diphosphate
$CO_2 + P_i$
DIPHOSPHO-MEVALONATE DECARBOXYLASE
HMG-CoA
trans-Methyl-glutaconate shunt
ISOPENTENYL-DIPHOSPHATE ISOMERASE
3,3-Dimethylallyl diphosphate
Isopentenyl diphosphate
Isopentenyl tRNA
CIS-PRENYL TRANSFERASE
PP_i
Geranyl diphosphate
Prenylated proteins
TRANS-PRENYL TRANSFERASE
Side chain of ubiquinone
Heme a
Farnesyl diphosphate
Dolichol
SQUALENE SYNTHETASE
$NADPH + H^+$
Mg^{2+}, Mn^{2+}
$NADP^+$
$2PP_i$
Squalene


Mevalonate
ATP
ADP
Mevalonate kinase
5-Phosphomevalonate
ATP
ADP
Phosphomevalonate kinase
5-Pyrophosphomevalonate
ATP
ADP
Pyrophosphomevalonate kinase
3-Phospho-5-Pyrophosphomevalonate
Pyrophosphomevalonate decarboxylase
CO_2 + Pi
Δ^3-Isopentenyl pyrophosphate
Isopentenyl pyrophosphate isomerase
Dimethylallyl pyrophosphate
Activated isoprenes

Stage 3 involves the polymerization of six 5-carbon isoprene units to form a 30 carbon linear structure of squalene

- Isopentenyl pyrophosphate and dimethylallypyrophosphate undergo a head-to-tail condensation, in which one pyrophosphate group is displaced and a 10 carbon chain , geranylpyrophosphate is formed (The ‘head’ is the end to which the pyrophosphate is attached)
- Geranylpyrophosphate undergoes undergoes another head-to-tail condensation with isopentenylpyrophosphate yielding the 15-carbon intermediate farnesyl pyrophosphate
- Finally two molecules of farnesyl pyrophosphate join head to head with the elimination of both pyrophosphate groups forming squalene
- Farnesyl pyrophosphate also gives rise to dolichol and ubiquinone
  - Dolichol takes part in glycoprotein synthesis by transferring carbohydrate residues to asparagine residue of the polypeptide
  - Ubiquinone is a member of the respiratory chain in the mitochondria


3,3-Dimethylally pyrophosphate
+
Isopentenyl pyrophosphate
Cis-prenyl transferase (head to tail condensation)
PP_i
Geranyl pyrophosphate
Isopentenyl pyrophosphate
Cis-prenyl transferase (head to tail condensation)
PP_i
Farnesyl pyrophosphate
Farnesyl pyrophosphate
NADPH + H^+
$NADP^+$
Squalene synthetase (head to headl condensation)
Mg^{2+}, Mn^{2+}
2 PP_i
Squalene

Finally, in stage 4, the cyclizationof squalene forms the four rings of the steroid nucleus, and a further series of changes (oxidations, removal or migration of methyl groups) leads to the final product cholesterol

Squalene

Squalene monooxygenase — NADPH + H^+, O_2 → H_2O, $NADP^+$

Squalene 2,3-epoxide

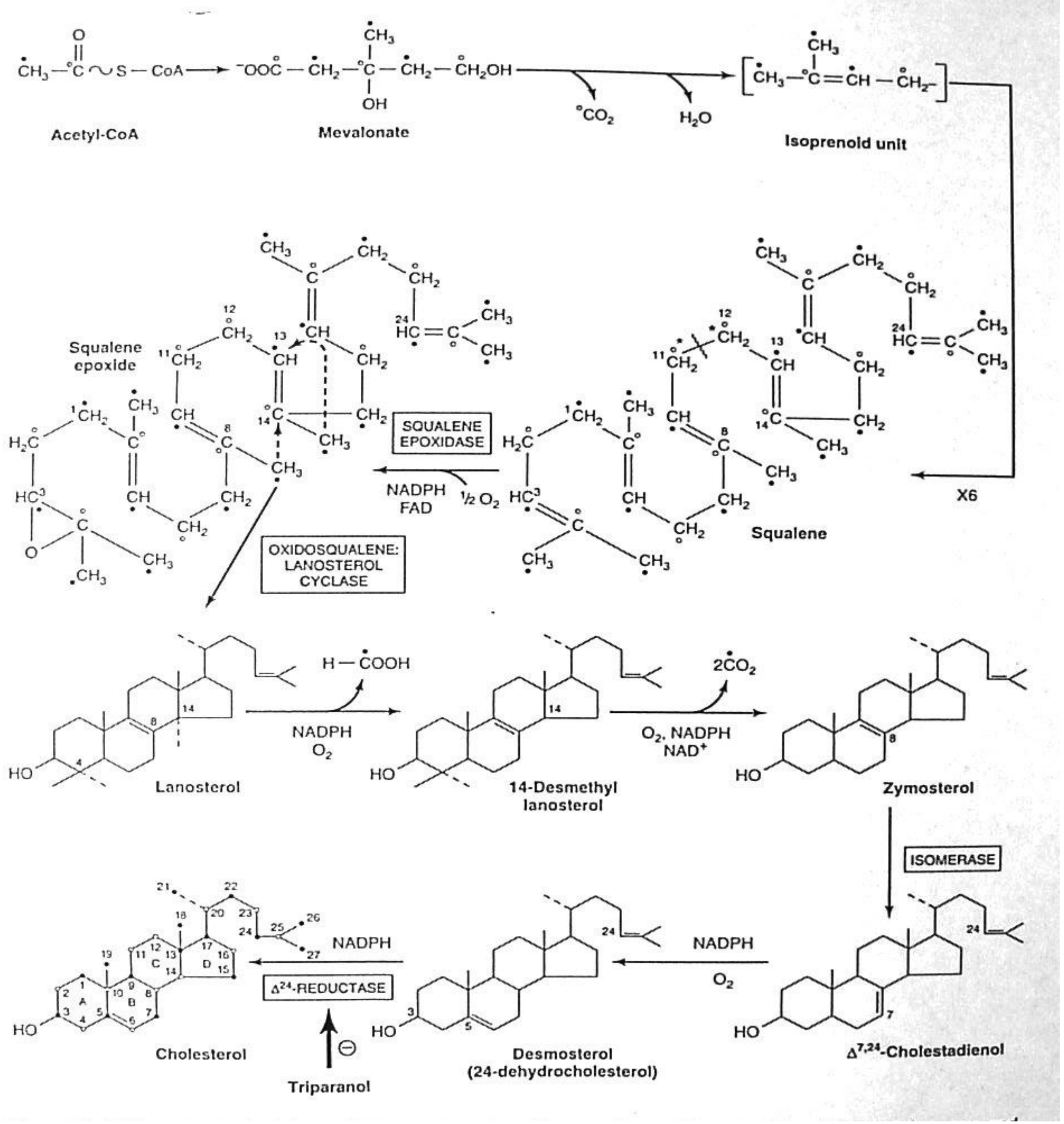
Cyclase

Lanosterol

Further steps

Cholesterol

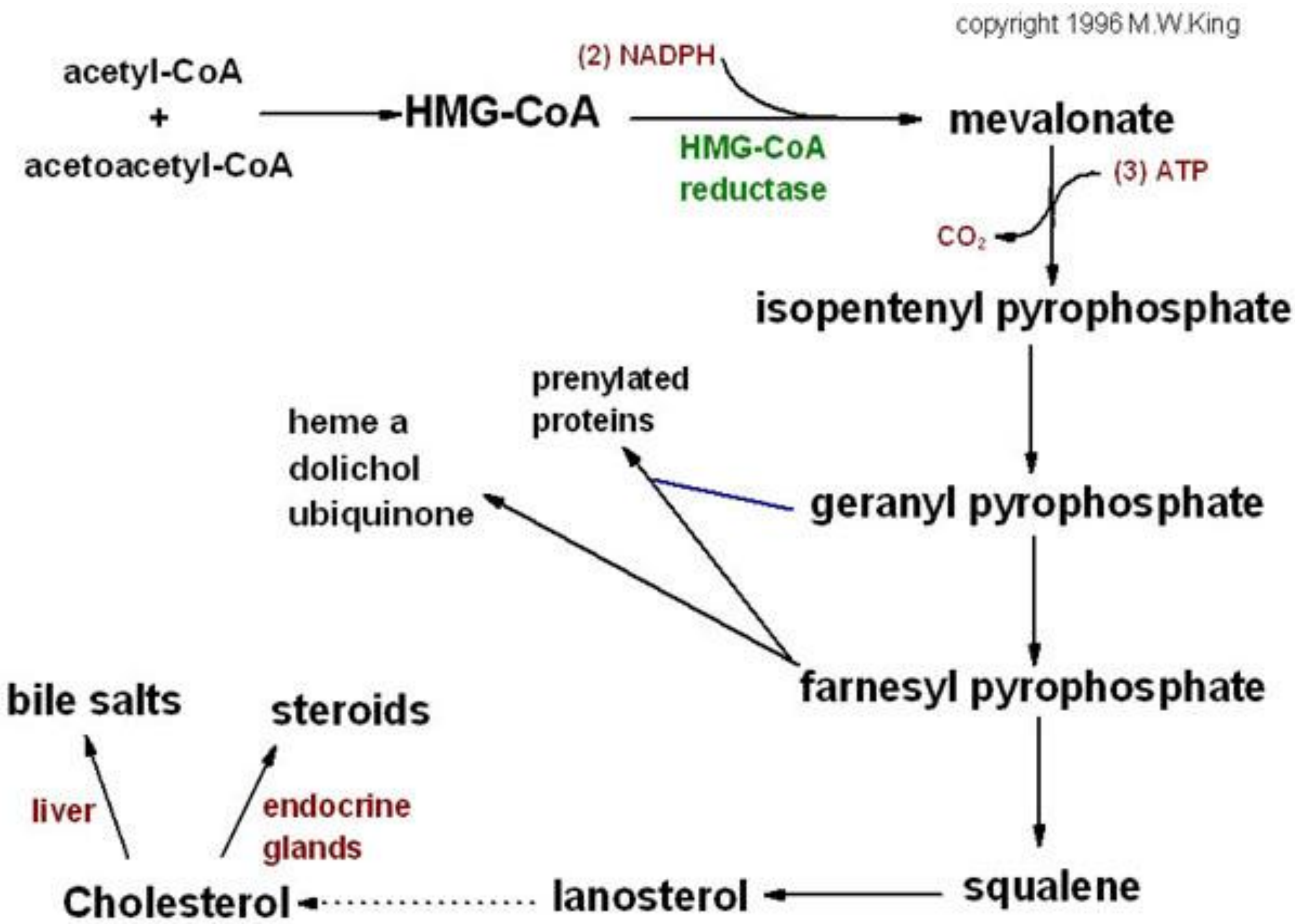
Before ring closure, squalene is converted to squalene 2,3-epoxide by a mixed-function oxidase in the endoplasmic reticulum. Thereafter, cyclization occurs under the action of the enzyme oxidosqualene: lanosterol cyclase.


Acetyl-CoA
Mevalonate
CO2
H2O
Isoprenoid unit
X6
Squalene
SQUALENE EPOXIDASE
NADPH
FAD
½ O2
Squalene epoxide
OXIDOSQUALENE: LANOSTEROL CYCLASE
Lanosterol
H — COOH
NADPH
O2
14-Desmethyl lanosterol
2CO2
O2, NADPH
NAD+
Zymosterol
ISOMERASE
Δ7,24-Cholestadienol
NADPH
O2
Desmosterol
(24-dehydrocholesterol)
NADPH
Δ24-REDUCTASE
Triparanol
Cholesterol

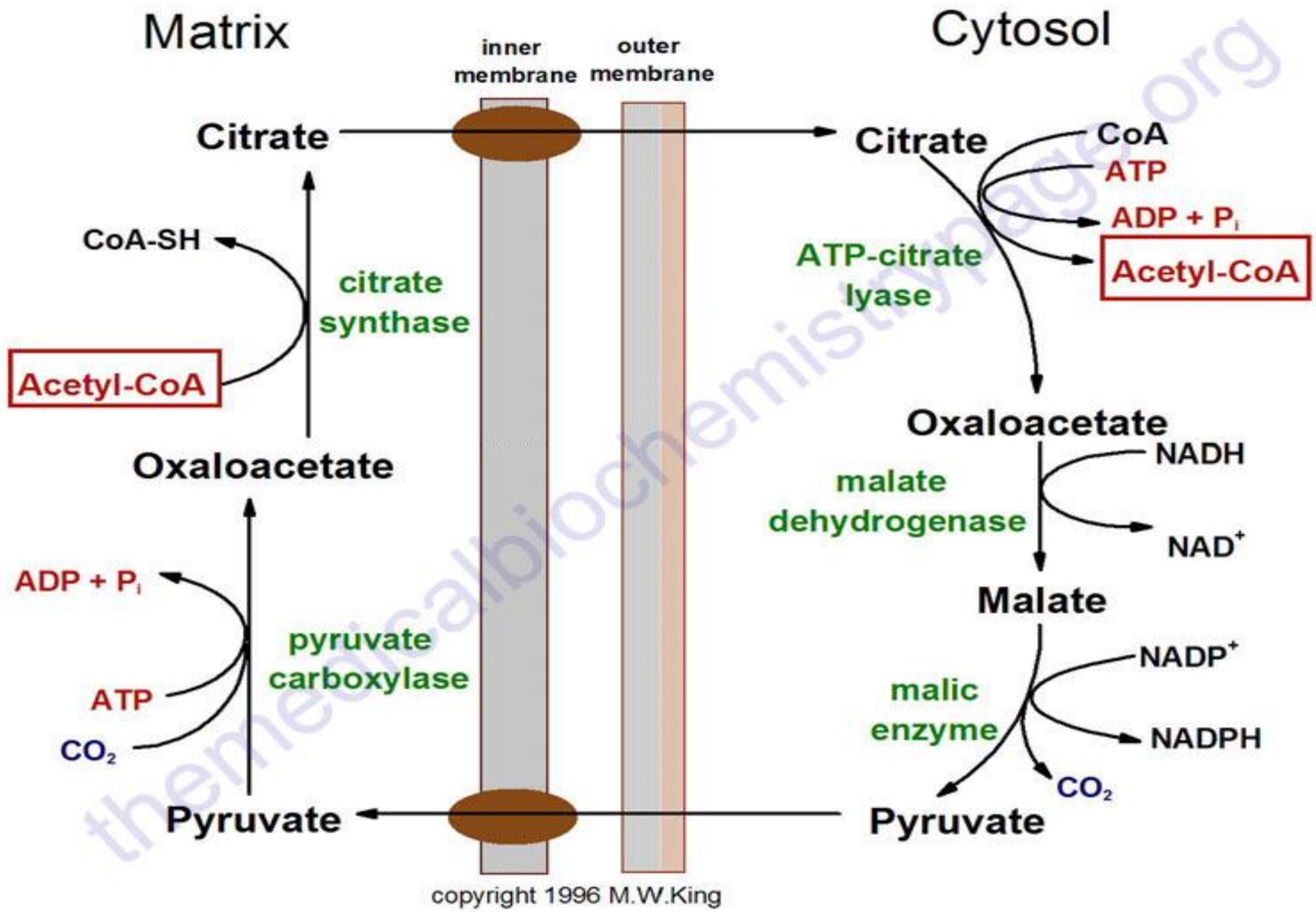
- Squalene folds into a structure that resembles a steroid nucleus
- Before ring closure squalene is converted to squalene 2,3-epoxide by *squalene monooxygenase* in the endoplasmic reticulum.
- The formation of cholesterol from lanosterol takes place in the endoplasmic reticulum and involves changes in the steroid nucleus and side chain
- Hepatic synthesis of cholesterol is at a peak at about 6 hours after dark and at a minimum some 6 hours after exposure to light.
  - This rhythm is the result of corresponding changes in HMG-CoA reductase activity.
  - Exploiting this, the drugs inhibiting cholesterol synthesis (statins) are usually taken at night to ensure maximal effect.

Why statins (Inhibitors of HMG CoA Reductase.) lower circulating cholesterol?

- If cells in peripheral tissues can't synthesize cholesterol, they respond by up-regulating their cell-surface LDL receptors to take in more chosterol from circulating LDL particles. This lowers circulating LDL cholesterol levels dramatically; Also lowers VLDL and IDL levels; Modestly increases HDL



Pathway of cholesterol biosynthesis. Synthesis begins with the transport of acetyl-CoA from the mitochondrion to the cytosol. The rate limiting step occurs at the 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA) reducatase, HMGR catalyzed step. The phosphorylation reactions are required to solubilize the isoprenoid intermediates in the pathway. Intermediates in the pathway are used for the synthesis of prenylated proteins, dolichol, coenzyme Q and the side chain of heme *a*.



Pathway for the movement of acetyl-CoA units from within the mitochondrion to the cytoplasm for use in lipid and cholesterol biosynthesis. Note that the cytoplasmic malic enzyme catalyzed reaction generates NADPH which can be used for reductive biosynthetic reactions such as those of fatty acid and cholesterol synthesis.

Regulation of Cholesterol Synthesis

- Cholesterol from both diet and synthesis is utilized in the formation of membranes and in the synthesis of the steroid hormones and bile acids. The greatest proportion of cholesterol is used in bile acid synthesis.

- The cellular supply of cholesterol is maintained at a steady level by three distinct mechanisms:
  - Regulation of HMGR activity and levels
  - Regulation of excess intracellular free cholesterol through the activity of acyl-CoA:cholesterol acyltransferase, ACAT
  - Regulation of plasma cholesterol levels via LDL receptor-mediated uptake and HDL-mediated reverse transport.
- Regulation of HMGR activity is the primary means for controlling the level of cholesterol biosynthesis.
- The enzyme is controlled by four distinct mechanisms: feed-back inhibition, control of gene expression, rate of enzyme degradation and phosphorylation-dephosphorylation.
  - The first three control mechanisms are exerted by cholesterol itself
  - The fourth through covalent modification occurs as a result of phosphorylation and dephosphorylation
    - Phosphorylation of the enzyme decreases its activity. HMGR is phosphorylated by AMP-activated protein kinase, **AMPK** (this is not the same as cAMP-dependent protein kinase, PKA).

- The activity of HMGR is additionally controlled by the cAMP signaling pathway.
- Insulin leads to a decrease in cAMP, which in turn activates cholesterol synthesis. Alternatively, glucagon and epinephrine, which increase the level of cAMP, inhibit cholesterol synthesis.
- Long-term control of HMGR activity is exerted primarily through control over the synthesis and degradation of the enzyme.
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The fate of cholesterol

- The sterol ring of cholesterol cannot be degraded in the human body. Therefore it is excreted either in the free form as the biliary cholesterol, or in the form of bile acids. Most of the bile acids are however returned to the liver after reabsorption in the terminal ileum. This is known as the enterohepatic circulation.
- Much of cholesterol synthesis takes place in the liver
  - A small fraction is incorporated into the membranes of hepatocytes
  - Most exported in one of the three forms
    - Biliary cholesterol
    - Bile acids and salts - hydrophilic cholesterol derivatives that aid in lipid digestion
    - Cholesterol esters – formed in the liver through the action of acyl-CoA cholesterol acyl transferase (ACAT). Transfers a fatty acid from acylcoenzyme A to the hydroxyl group of cholesterol, converting the cholesterol into a more hydrophobic form

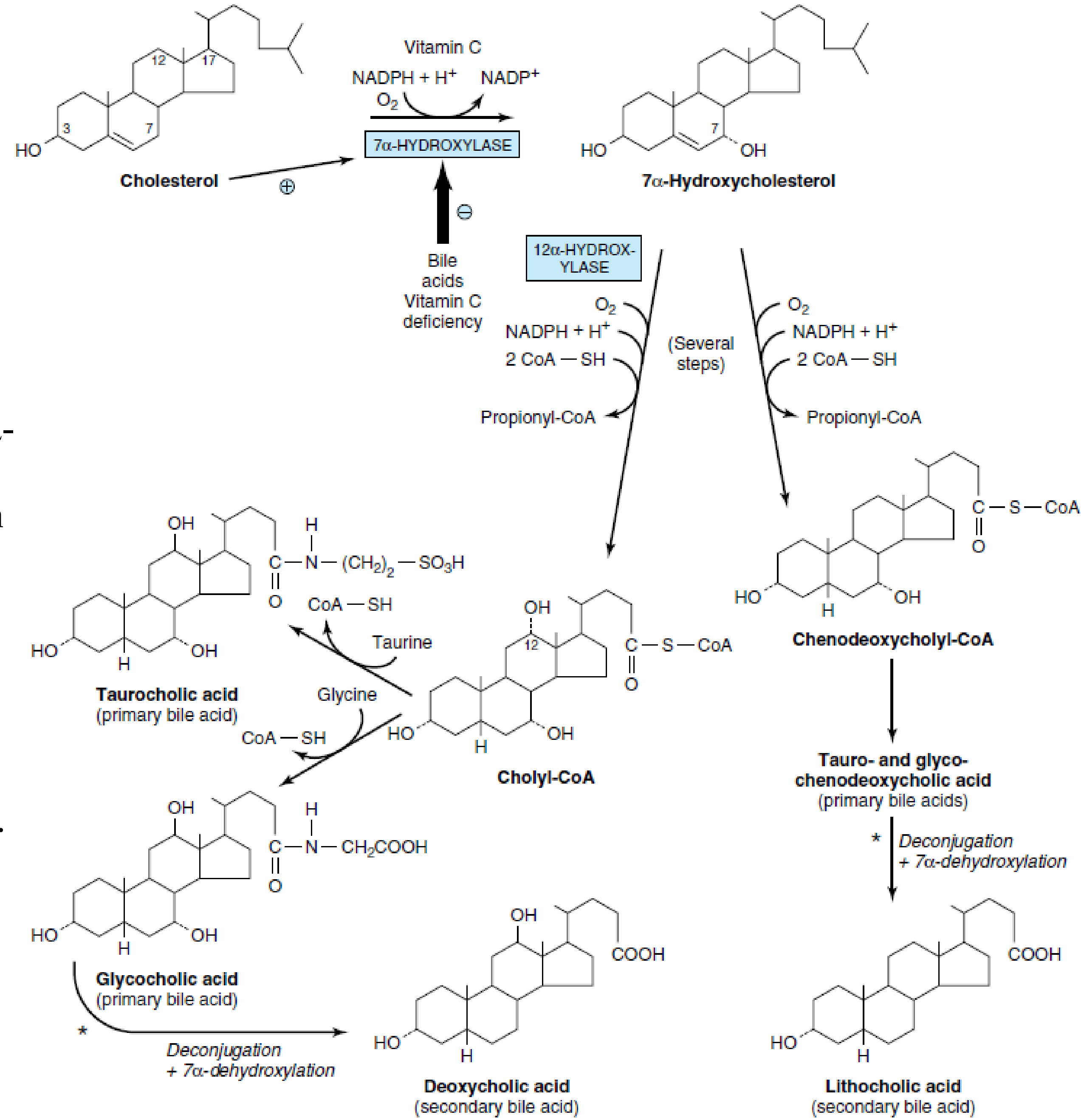
- Many factors are involved in the regulation of the intracellular concentration of cholesterol. Under normal circumstances there is an inverse relationship between dietary cholesterol intake and cholesterol biosynthesis. This ensures a relatively constant daily supply of cholesterol but it explains why dietary restriction is only likely to achieve a 15% reduction in circulating cholesterol concentrations.
- When the total amount of cholesterol synthesized and obtained in the diet exceeds the amounts required for synthesis of membranes, bile salts, and steroids pathological accumulation of cholesterol occurs- atherosclerosis
- Atherosclerosis is linked to high levels of LDL-bound cholesterol (bad cholesterol)
- There is also a negative correlation between HDL-bound cholesterol (good cholesterol) levels and arterial disease
  - HDL act as scavengers of cholesterol
- In familial hypercholesterolemia blood levels of cholesterol are very high and afflicted individuals develop severe atherosclerosis in childhood
  - LDL receptor is defective in these individuals and the receptor-mediated uptake of cholesterol carried by LDL does not occur. Therefore cholesterol is not cleared from the blood
    - Two products derived from fungi (lovastatin and compactin) are used to treat patients with familial hypercholesterolemia
    - The two compounds and several synthetic analogues are competitive inhibitors of HMG-CoA reductase

Bile acids

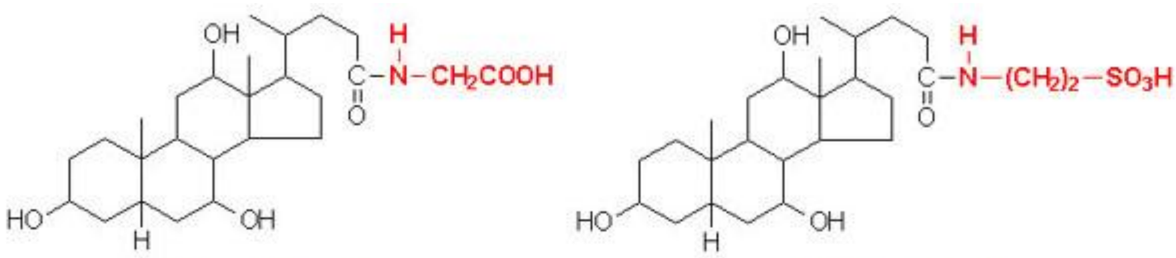
- All have 24 carbon atoms
- Have a saturated steroid nucleus. Differ in the number and position of hydroxyl groups.
- All hydroxyl groups are in the α-configuration (below the plane of the ring)
- Microsomal 7 α–hydroxylase converts cholesterol to 7-hydroxycholesterol
  - Rate limiting step
  - Introduces a hydroxyl group at the 7 α– position
  - Microsomal monooxygenase with cytochrome p 450 and requires NADPH and O_2
- An activating enzyme converts 7-hydroxycholesterol to bile acids which exist within liver cells as colyl-CoA and chenodeoxycholyl-CoA derivatives
- Another enzyme catalyzes conjugation of activated CoA derivatives with glycine or taurine to form glycocolic acid or glycochenodeoxycholic acid and taurocholic acid or taurochenodeoxycholic acid – these are the primary bile acids
- In humans ratio of glycine to taurine conjugates is 3:1
- At physiological pH the bile acids are mainly ionized and so occur as Na^+ or K^+ salts – the terms bile acids and bile salts are sometimes used interchangeably

Bile acid synthesis and utilization

Synthesis of the 2 primary bile acids, cholic acid and chenodeoxycholic acid. The reaction catalyzed by the 7α-hydroxylase is the rate limiting step in bile acid synthesis. Conversion of 7α-hydroxycholesterol to the bile acids requires several steps not shown in detail in this image. Only the relevant co-factors needed for the synthesis steps are shown.

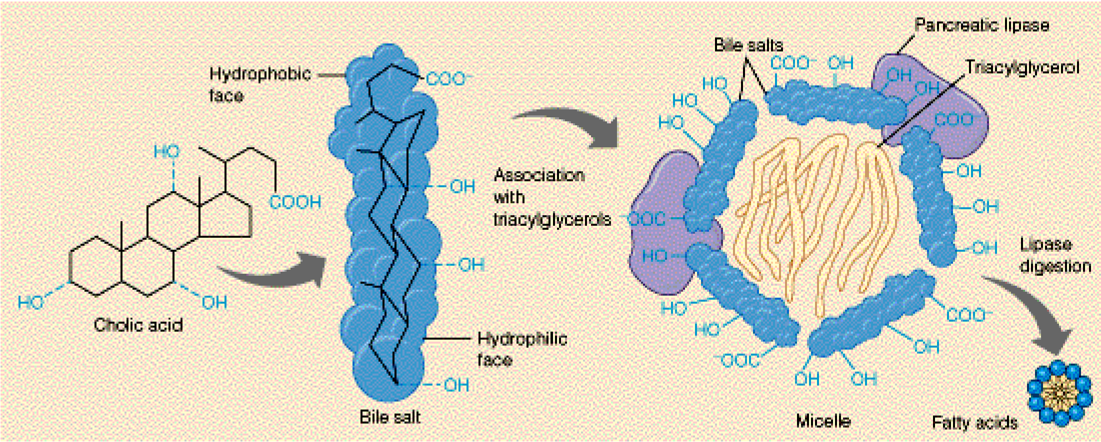


- Deoxycholic and lithocholic bile acids are formed within the intestine through action of bacterial enzymes
  - There is hydrolysis of the amide linkage of taurine and glycine
  - Removal of 7-hydroxyl group
- Up to 30g of bile acids per day enter the small intestine and only about 2% of these are lost in feces
  - Passive reabsorption of bile acids occurs in the ileum and colon; active reabsorption in the ileum
  - Noncovalently bound to albumin in portal vein - to the liver where reexcretion takes place --enterohepatic circulation
- Cholestyramine
- Interferes with enterohepatic circulation
  - Activates 7-hydroxylase
  - Increases bile acid synthesis
  - Increases bile acid excretion



Glycocholic acid Taurocholic acid

- Vitamin C deficiency interferes with bile acid formation at 7 hydroxylation step. Leads to cholesterol accumulation hypercholesterolemia …



Hydrophobic face
HO
COOH
HO
OH
Cholic acid
COO⁻
OH
OH
Hydrophilic face
OH
Bile salt
Association with triacylglycerols
Bile salts
Pancreatic lipase
Triacylglycerol
Lipase digestion
Micelle
Fatty acids

- Bile acids are involved in four primary significant functions:
  - their synthesis and subsequent excretion in the feces represent the only significant mechanism for the elimination of excess cholesterol.
  - bile acids and phospholipids solubilize cholesterol in the bile, thereby preventing the precipitation of cholesterol in the gallbladder.
  - they facilitate the digestion of dietary triacylglycerols by acting as emulsifying agents that render fats accessible to pancreatic lipases.
  - they facilitate the intestinal absorption of fat-soluble vitamins.

TASK

- A 35-year-old woman was seen in the emergency room because of recurrent abdominal pain. The history revealed a 2-year pattern of pain in the upper right quadrant, beginning several hours after the ingestion of a meal rich in fried/fatty food. Ultrasonographic examination demonstrated the presence of numerous stones in the gallbladder. The patient initially elected treatment consisting of exogenously supplied chenodeoxycholic acid, but eventually underwent surgery for the removal of the gallbladder, and had a full recovery. The rationale for the initial treatment of this patient with chenodeoxycholic acid is that this compound:

A. interferes with the entero-hepatic circulation.

B. inhibits cholesterol synthesis.

C. increases de novo bile acid production.

D. increases cholesterol solubility in bile.